Computer Network I

Reti di Calcolatori I

Università di Napoli Federico II – Scuola Politecnica e delle Scienze di Base
Corso di Laurea in Informatica

Riccardo Caccavale

(riccardo.caccavale@unina.it)



UNIVERSITÀ DEGLI STUDI DI NAPOLI
FEDERICO II



DIETI

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Overview

About the Course

- Computer networks I (Reti di calcolatori I):
  - CFU: 6
  - Modality: Frontal lessons and class examples (in Italian).
  - Team: oryjf76.
    - Communications will be mostly on Teams

- Evaluation:
  - Oral examination.
  - Dates depend on the calendar.

- Text books:
  - James Kurose, Keith Ross, Computer Networking A Top-Down Approach. [Main reference]
  - Andrew Tanenbaum, David Wetherall, Computer Networks.

- Additional Materials:
  - Slides (in English).

Overview
About Myself



- Riccardo Caccavale, PhD
  - Ricercatore (RTD-A) presso il dipartimento di Ingegneria Elettrica e Tecnologie dell'Informazione (DIETI).
    - E-mail: riccardo.caccavale@unina.it
    - Web (UNINA): https://www.docenti.unina.it/riccardo.caccavale
    - Web (wpage): http://wpage.unina.it/riccardo.caccavale/index.html
    - Workplaces:
      - Via Claudio 21, building 3/A, room 2.11
      - Via Claudio 21, building 5/A, PRISMA Lab
  - Research activities in the field of robotics, especially in AI-based, cognitive, and autonomous robotics.
  - Member of the PRISMA Lab (from 2016).
    - Website: https://prisma.dieti.unina.it/index.php

Overview
Schedule and Dates

- Class Schedule:

Giorno	Ora	Luogo
Lunedì	8:30-10:30	Via Claudio 21, aula CL-II-3
Martedì	8:30-10:30	Via Claudio 21, aula CL-II-2

- Assistantship (Ricevimento):

Giorno	Ora	Luogo
Lunedì	10:00 – 11:00	Teams - Via Claudio 21, edificio 3/A, stanza 2.11
	10:30-11:30	Possibly on appointment

Overview

Motivations

- The human being is a social animal living in a connected world.
  - Around 66% of the world's population connected to Internet [Fobes, 2024].
  - Internet users spend an average of 400 minutes (nearly 7 hours) a day online [Fortune, 2024].
- Everyday activities increasingly rely on connection.
  - Watching TV, driving, exchanging money, paying taxes, etc.
- A lot of todays commercial applications (software) and devices (hardware) use Internet to run.
  - Globalization/optimization of resources.
- The acronym ICT (Information and Communication Technologies) stresses the central role of communication in computer science.



Overview

Objectives

- To introduce fundamental elements on which computer networks (in general) and internet (in particular) are based.
- To provide basic concepts and tools allowing students to understand, analyze and suitably design computer networks and network applications.
- To introduce and understand network protocols, having knowledge about networks infrastructure and devices.
- To understand potentialities, as well as risks, limitations and security issues of computer networks.

Overview

Syllabus

- Introduction to computer networks: goals and applications, history of internet, main concepts, network topologies and types, services (ISPs, grid, cloud, IoT), standards (layered models, IOS/OSI and reference models, protocols).
- Application layer: client-server ad peer-to-peer architectures, FTP and SSH protocols (basis), structure of URL, HTTP protocol, mailing protocols (STMP, POP3, IMAP), DNS, BitTorrent protocol.
- Transport layer: fundamentals, UDP protocol, reliable data transfer, TCP protocol.
- Network layer: fundamentals, routers, IPv4 protocol, DHCP, NAT, IPv6, families of routing algorithms.
- Access layer (data link + physical): network interfaces, parity check and CRC, collisions, ARP protocol, Ethernet protocol (basis).
- Introduction to network security: attacks on networks, introduction to cryptography, message integrity, AP and TLS/SSL, operational security (firewall, IDS).
- Network programming: UDP and TCP sockets, REST applications, data exchange formats.

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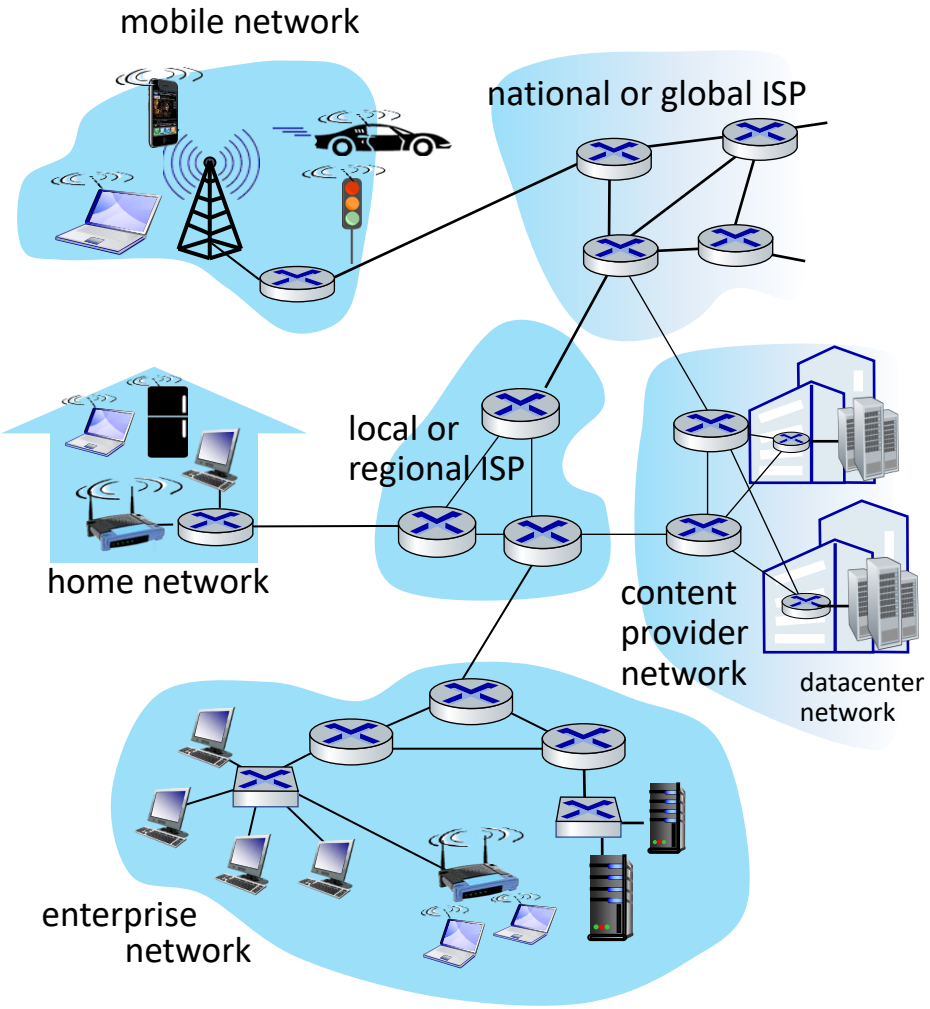
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Introduction

Computer Networks and Internet

Computer networking: the process of connecting computer together so that they can share information.

[Cambridge Dictionary]



- **Internet** (the most important computer network in the world) is probably the **largest engineered system ever created**.
  - Internet includes hundreds of millions of network **devices**, **links**, **computers**, etc. offering hundreds of **services** for the users.

- There are also **smaller networks** (local or detached from internet).

Introduction

History of Internet

- The **Internet** is the largest computer network, connecting devices all around the globe.



Internet is a network of networks. It is technically a huge *Wide Area Network* (WAN) that allows sub-networks and devices to be connected over different nations and continents.

The idea of such “universal” big network was initially theorized by researchers (most notably J.C.R. Licklider) around 1950 and started to become reality almost 10 years later.

Introduction

History of Internet

- The **Internet** is the largest computer network, connecting devices all around the globe.

- Brief history of Internet:
  - The Advanced Research Projects Agency (ARPA) of the U.S. Department of Defense awarded contracts for the development of the **ARPANET** project (1969).



Initial ARPANET having 3+1 nodes: SRI, UCLA, UCSB, and UTAH (1969)

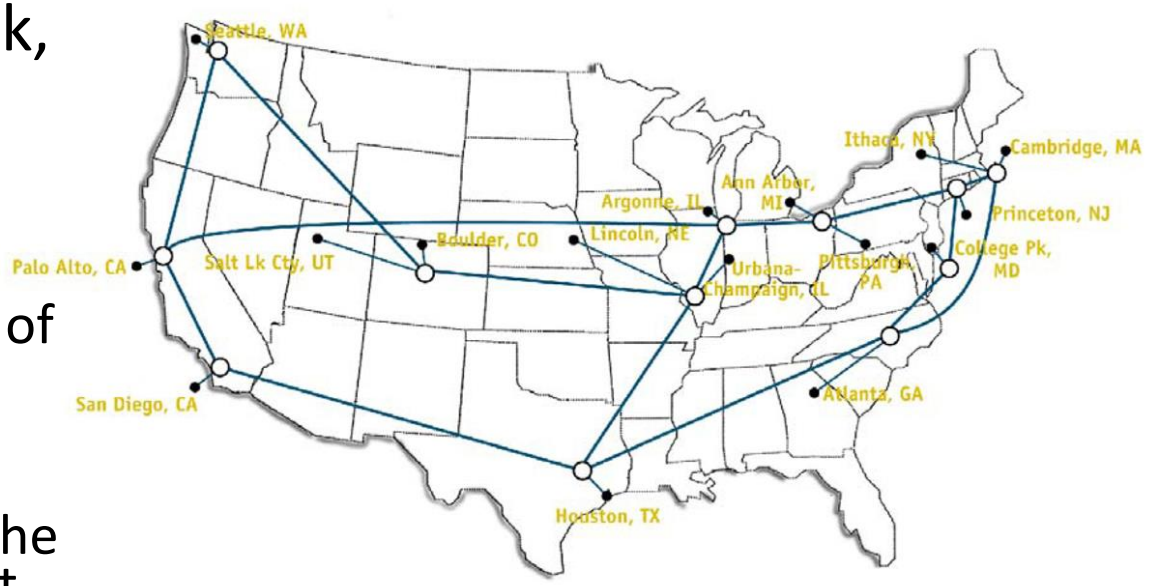
The initial ARPANET was mainly a closed network designed to include universities and research centers. Similar networks were independently created all around US and Europe.

Introduction

History of Internet

- The **Internet** is the largest computer network, connecting devices all around the globe.

- Brief history of Internet:
  - The Advanced Research Projects Agency (ARPA) of the U.S. Department of Defense awarded contracts for the development of the **ARPANET** project (1969).
  - R. Kahn (ARPA) and V. Cerf (Stanford) designed the Transmission Control Protocol (TCP) and Internet Protocol (IP), two protocols of the **Internet protocol suite** (1974).
  - The National Science Foundation (NSF) awarded contracts for the **NSFNET** project, a TCP-IP based network (1986).



NSFNET in 1992 (Wikipedia).

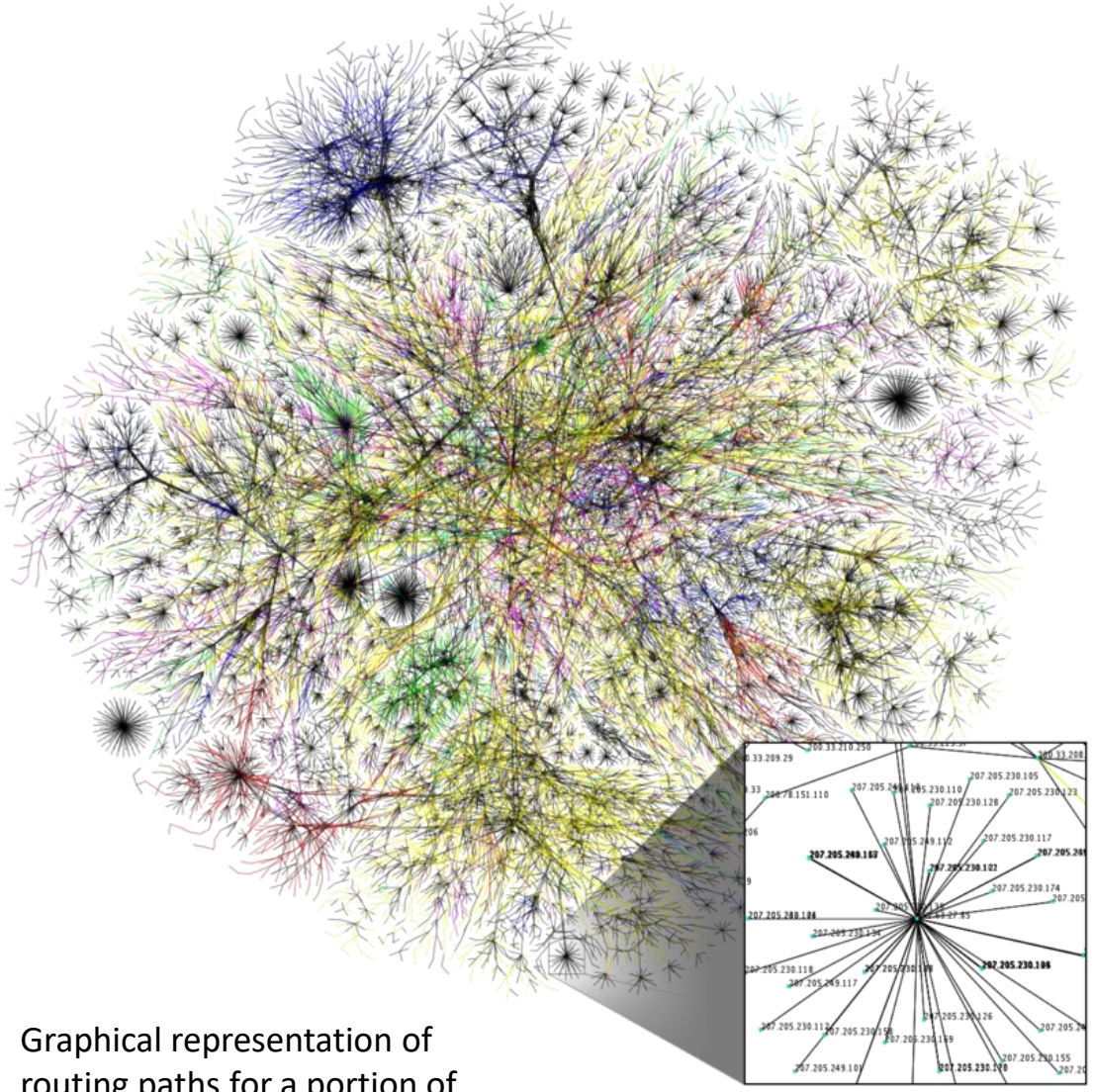
The NSFNET and the TCP/IP protocol went toward the idea of a *network of networks,* but both ARPANET and NSFNET carried **commercial restrictions**.

Introduction

History of Internet

- The **Internet** is the largest computer network, connecting devices all around the globe.

- Brief history of Internet:
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  - The National Science Foundation (NSF) awarded contracts for the **NSFNET** project, a TCP-IP based network (1986).
  - The **ARPANET decommissioned** (1990).
  - The **NSFNET decommissioned**, removing the last restrictions on the use of the Internet for commercial traffic (1995).

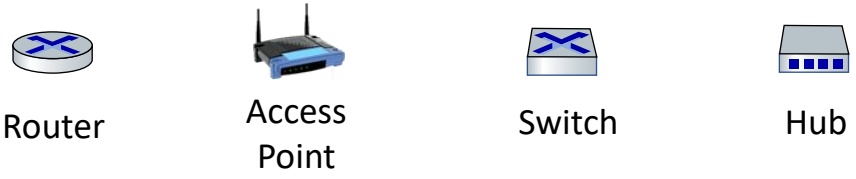


Graphical representation of routing paths for a portion of Internet (Wikipedia).

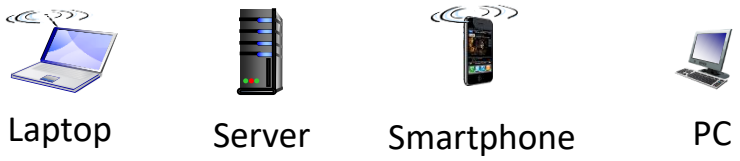
Introduction

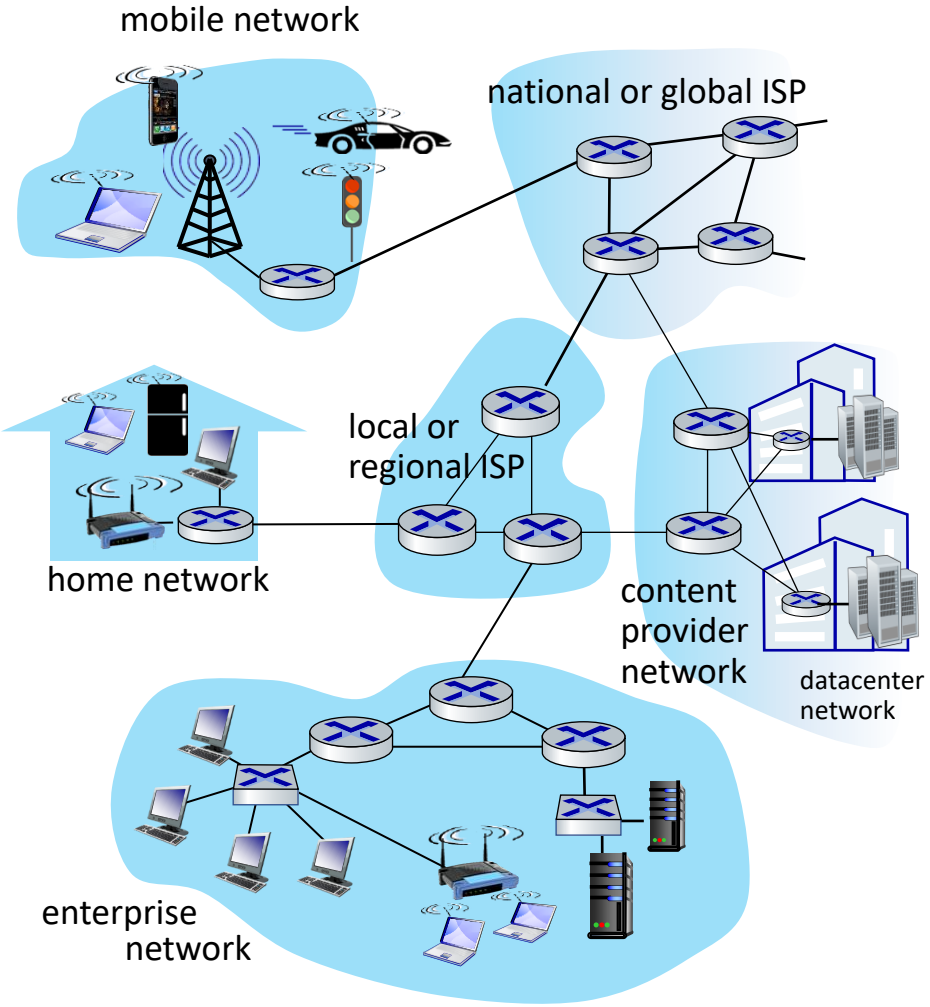
Networks Components: Devices, Links, Hosts

- Network **devices** and **links** are the **infrastructure** that allow hosts to connect:



- **Hosts** are devices on which **applications** (or programs) runs:



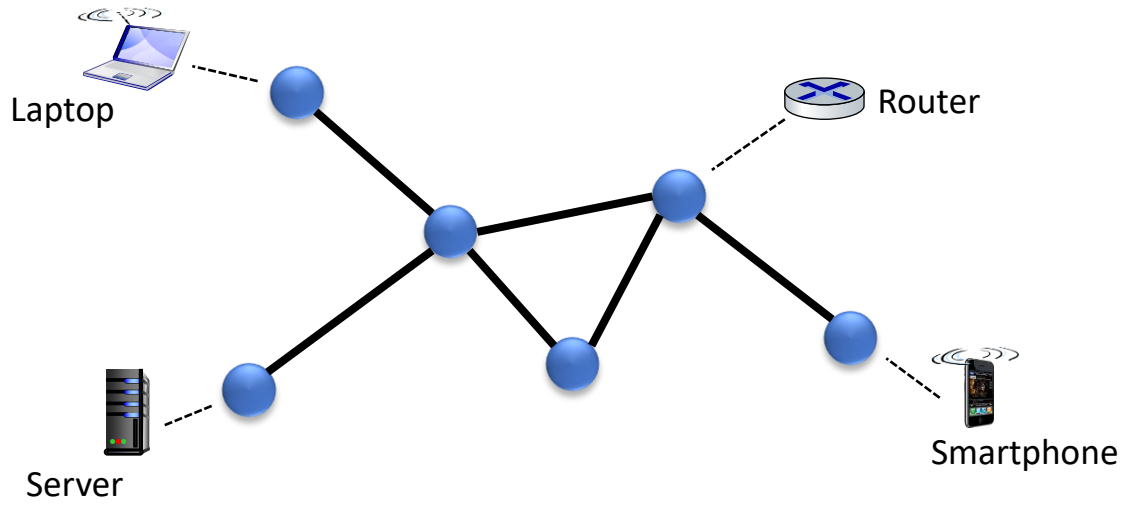


Introduction

Computer Networks and Graph Theory

- Computer networks share terminologies with graph theory:
  - The devices connected through the network are called **nodes** while the connections between nodes are called communication **links** (or **channels**).
  - A **sequence** of nodes/links is called **path**.

- The end-points (or end-systems) of the network, which provide or use **services**, are special nodes called **hosts**.



Ideally hosts are the leaves of the network (often it is not true) while intermediate nodes are routing devices (routers, switches).

Introduction
Data Communication

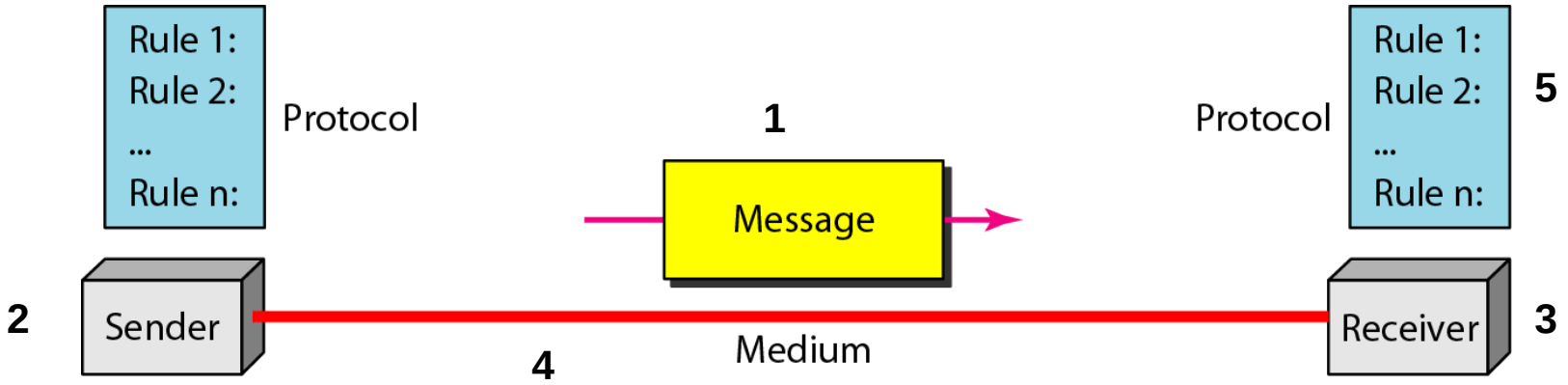
- The **goal of networking** is to share information: to allow large-scale internet functionalities as well as small-scale network functionalities we need to connect two hosts so they can **communicate data over distance**.

- **Data communication** is the process of exchanging data between two devices via some form of transmission medium (e.g., wires or wireless), it is often a **tradeoff** between:
  - **Reliability**: data is received correctly.
  - **Performance**: data is received within a reasonable amount of time.

- The **telecommunication** (i.e., communication at a distance) is the field that studies how to communicate data over distance.

Introduction
Data Communication

- Data communication has basically **five** components:
  1. **Message**: contains the **data** we want to communicate.
  2. **Sender**: the entity which is sending the message.
  3. **Receiver**: the entity which is supposed to receive the message.
  4. **Medium**: the channel between **sender** and **receiver** where the **message** travels.
  5. **Protocol**: a set of rules, known to **sender** and **receiver** used to manage the **message**.

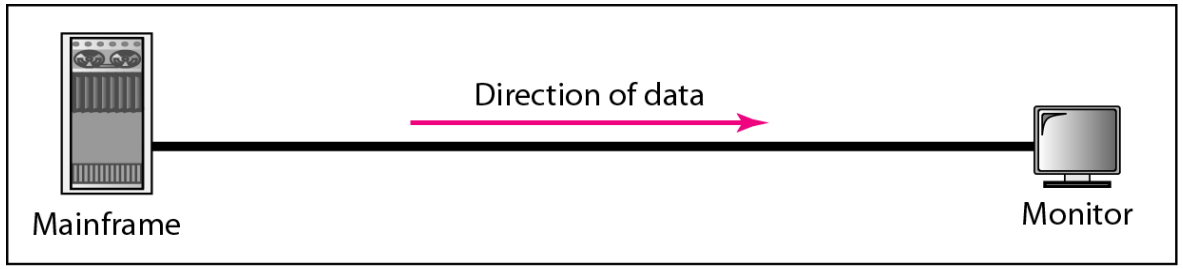


Introduction

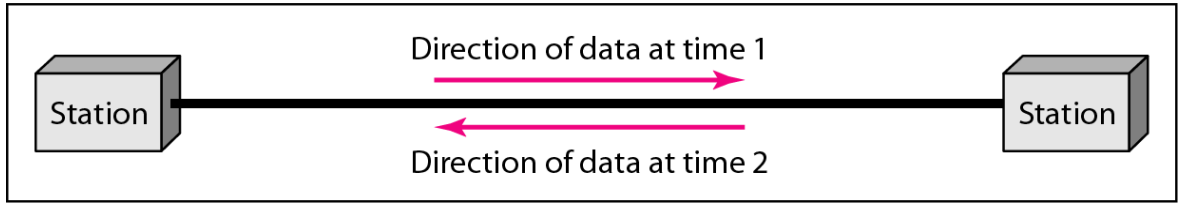
Data Representation and Flow

- The **data** we want to exchange can be represented in different forms.
  - Text, numbers, images, audio, video, etc.

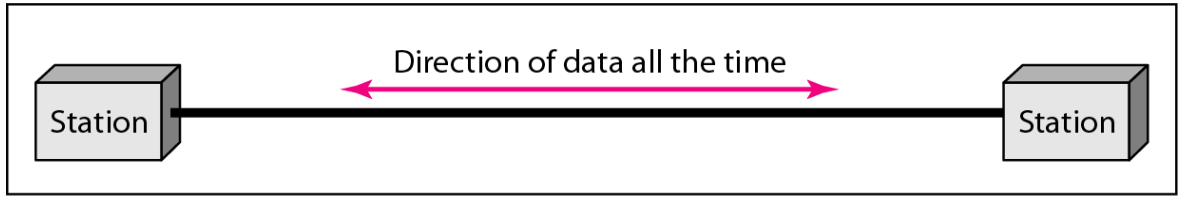
- Depending on the type and the purpose of the communication, the **data flow** can be:
  - **Simplex**: monodirectional.
  - **Half-duplex**: bidirectional (taking turns).
  - **Full-duplex**: bidirectional (simultaneous).



a. Simplex



b. Half-duplex



c. Full-duplex

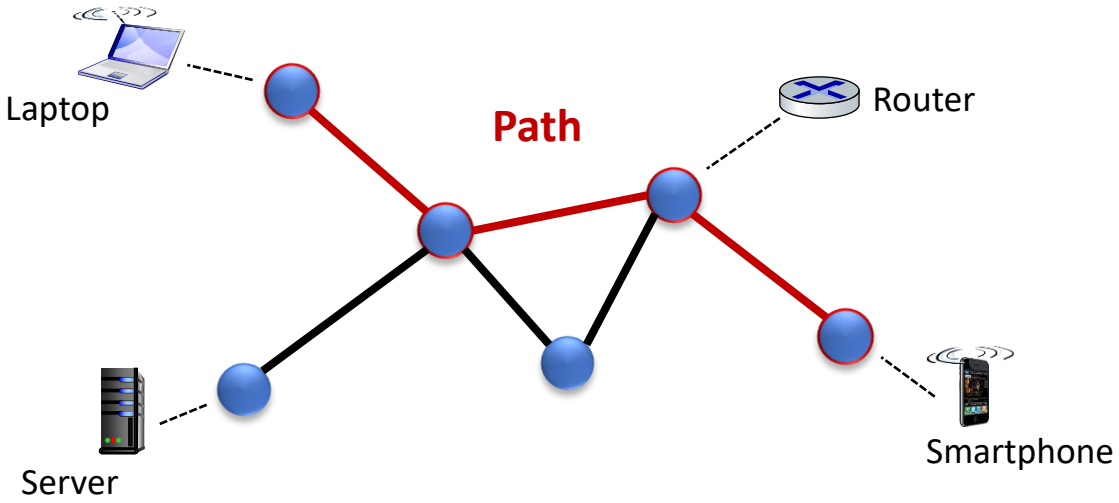
Introduction

Data Representation and Flow

- **Transmission rate** typically refers to the **maximum** amount of information that a **device** can transmit, measured in bits/secs (or bytes/secs).

- The **bandwidth** is the **maximum** amount of information that a **path** (links + nodes) can transmit, measured in **bits/secs** (or bytes/secs).

- The **throughput** is the **actual** (instantaneous) amount of information that is being transmitted over a **path**, measured in **bits/secs** (or bytes/secs).

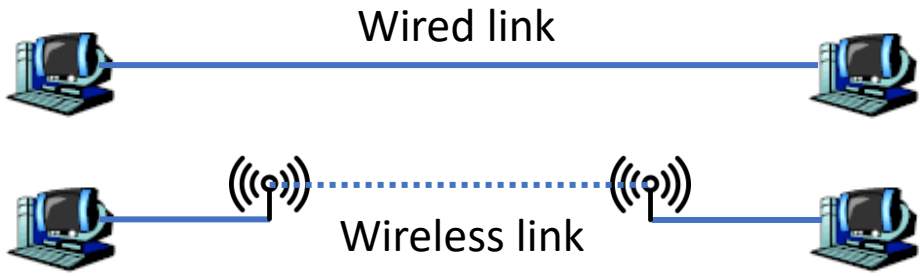


Introduction

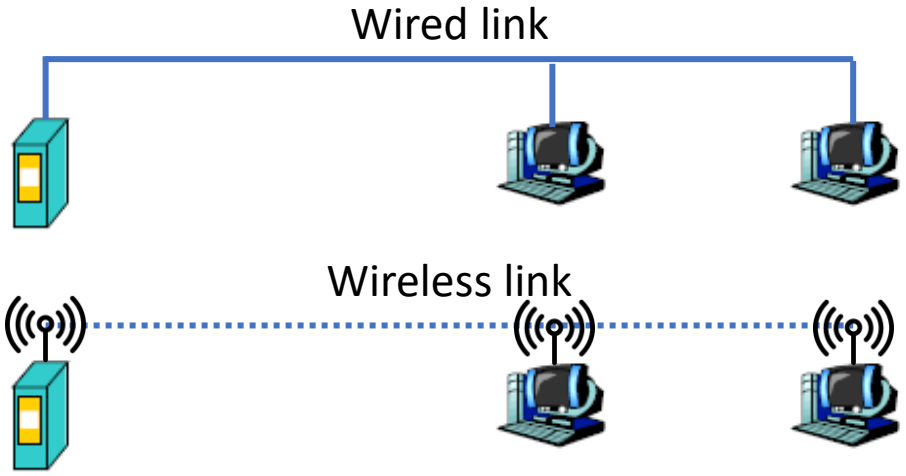
Types of Connections

- The **hosts** of a network can be connected in different ways.

- **Point-to-point**: a dedicated link is provided between two devices (wireless or wired).



- **Multipoint** (**broadcast**): more than two specific devices share a single link (wireless or wired).

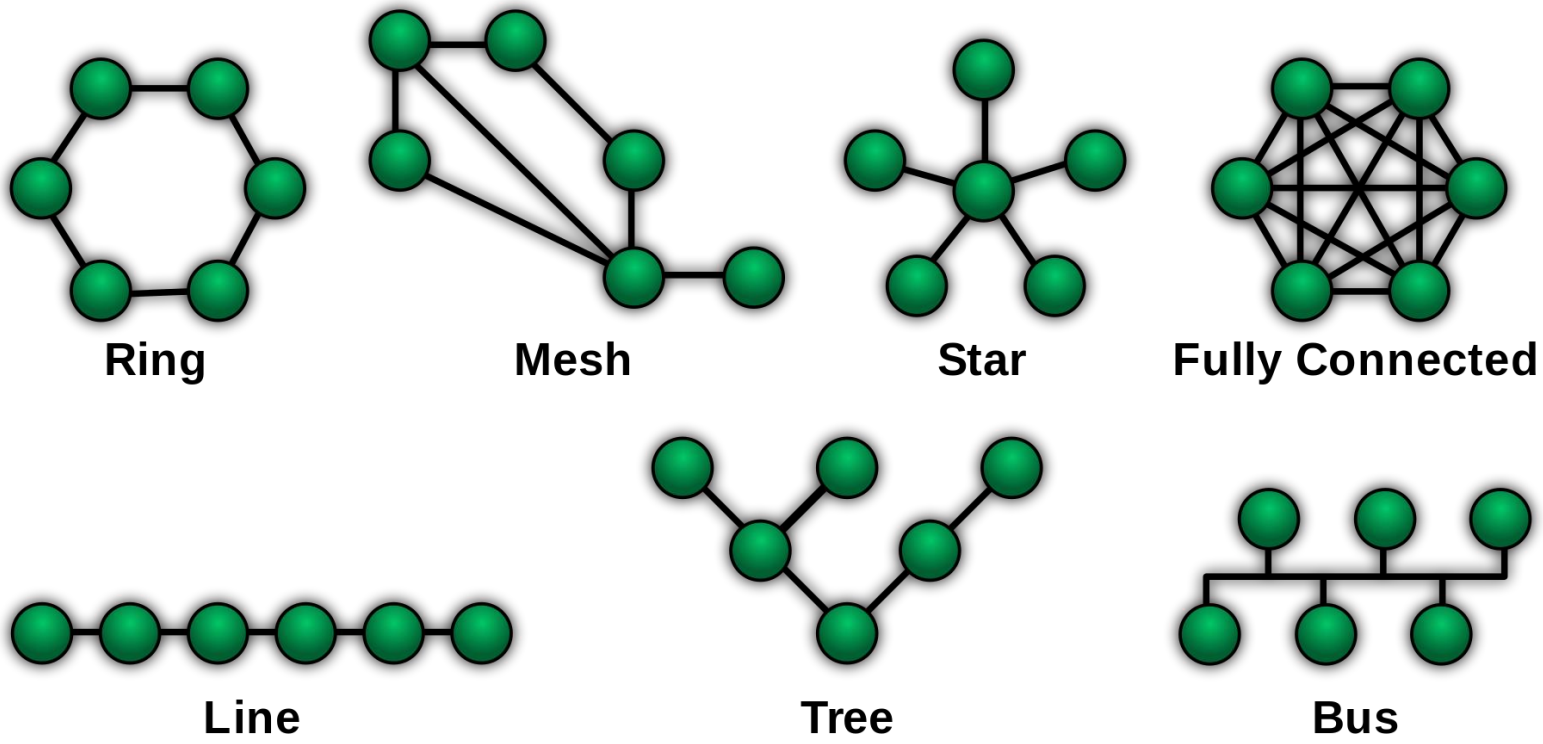


Introduction

Network Topologies

- There are different ways of connecting devices.

- The **network topology** is the arrangement of the elements (links, nodes, etc.) of a communication network.

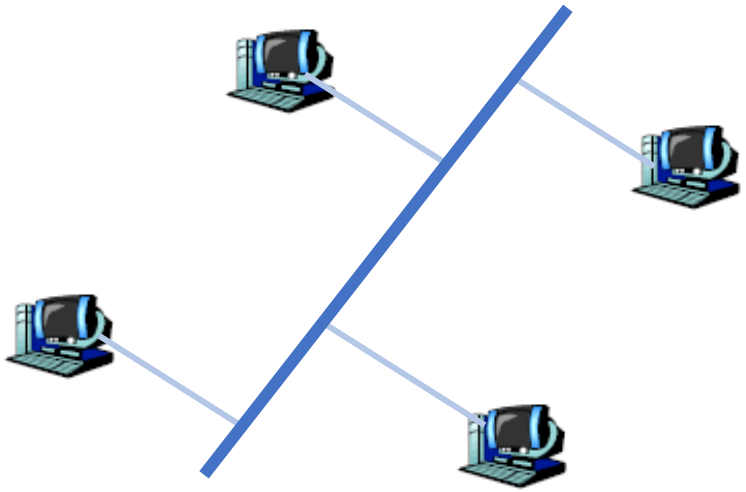


en.wikipedia.org

Introduction

Network Topologies: Bus

- In the **Bus** topology hosts are connected to a central backbone (bus) cable.
  - Messages sent by 2 hosts generate **collisions**.



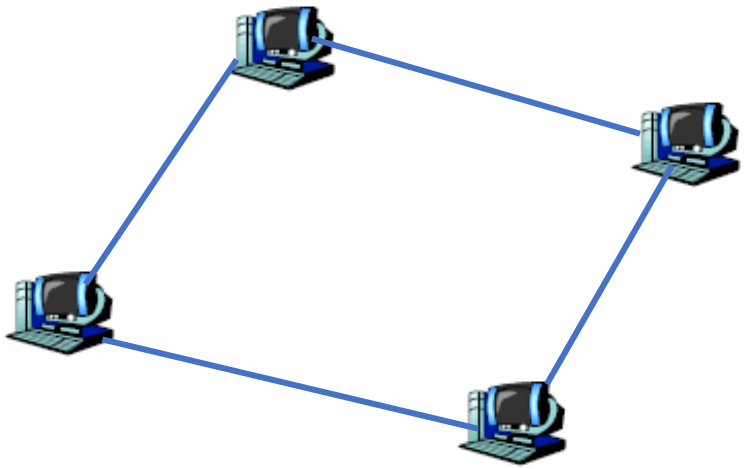
- Pros:
  - Simple and cheap.
  - Good for small networks.
- Cons:
  - Single point of failure (broken bus) but sub networks may still be available.

Bus networks have *1* physical duplex link (or *1* backbone and *n* links).

Introduction

Network Topologies: Ring

- In the **Ring** topology hosts are point-to-point connected to exactly two other ones. The signal is **forwarded** along the ring, from device to device, until it reaches its destination.



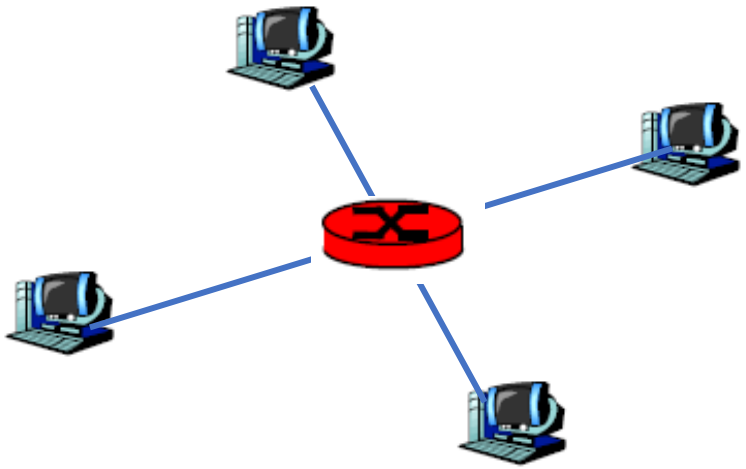
- Pros:
  - Simple and cheap.
  - Performs better than the bus.
- Cons:
  - Adding new nodes is harder.
  - Nodes malfunctioning may impair the network.

Ring networks have *n* duplex links.

Introduction

Network Topologies: Star

- In the **Star** topology hosts are linked to a central controller (hub, switch or router), there is no direct link between hosts.
- The central controller redirects messages.



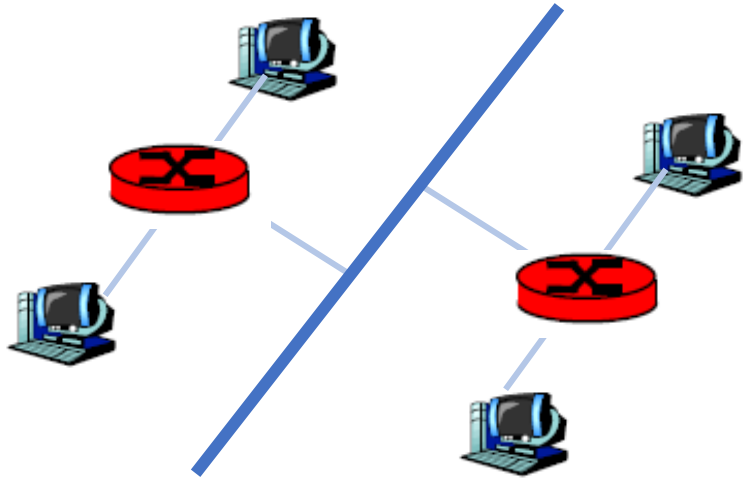
- Pros:
  - Less expensive, simple, robust, more scalable.
- Cons:
  - The controller must be reachable by all hosts.
  - Single point of failure.

Star networks have *1* controller and *n* physical duplex links!

Introduction

Network Topologies: Tree

- In the **Tree** topology multiple star topologies are integrated typically through a bus cable.

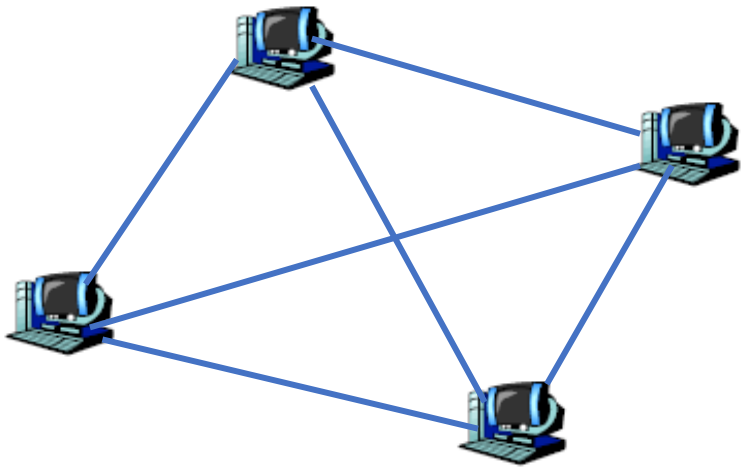


- Pros:
  - Versatile, scalable, robust.
  - Well supported by HW and SW providers.
- Cons:
  - Hard to configure.
  - Weakness of the bus.

Introduction

Network Topologies: Mesh

- In the **Mesh** topology hosts are point-to-point linked in a non-hierarchical way.
  - **Full mesh**: all nodes connected (full-connected).
  - **Partial mesh**: nodes connected with some others.



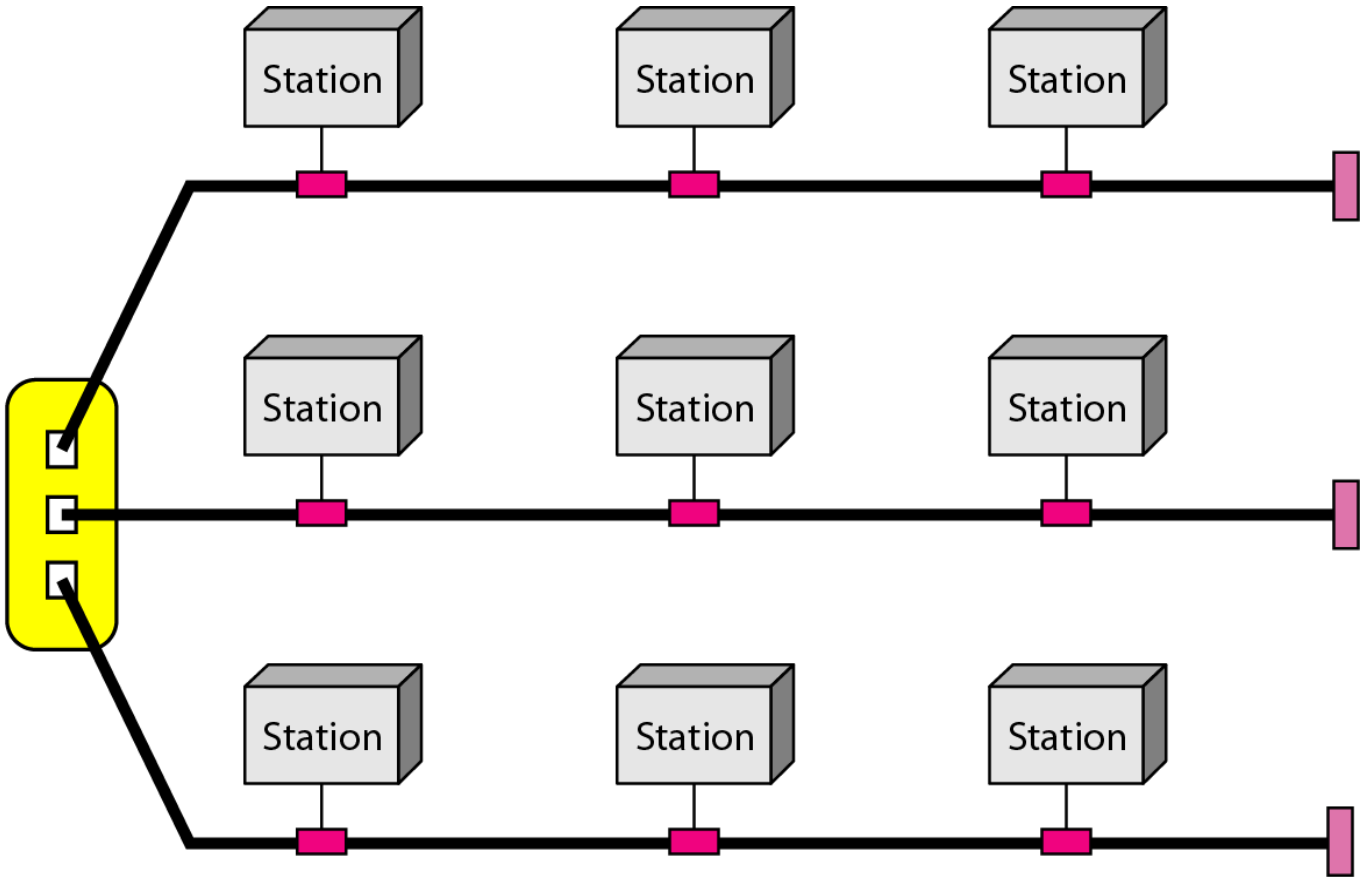
- Pros:
  - Low traffic, robust, secure, dedicated.
- Cons:
  - Hardly scalable.
  - Expansive (need for devices with multiple ports).

Full mesh networks have ${}^{n}C_{2} = n(n-1)/2$ physical duplex links!

Introduction

Network Topologies: Hybrid

- Topologies can be **mixed** into a hybrid network.

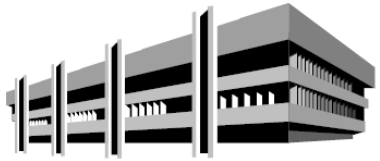


Star-backbone with 3 bus networks

Introduction

Categories of Networks

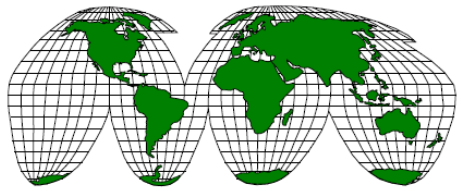
- Networks are categorized by dimension, number of hosts, bandwidth.
  - **Local Area Network** (LAN) or Wireless Local Area Network (WLAN).
  - **Metropolitan Area Network** (MAN).
  - **Wide Area Network** (WAN)



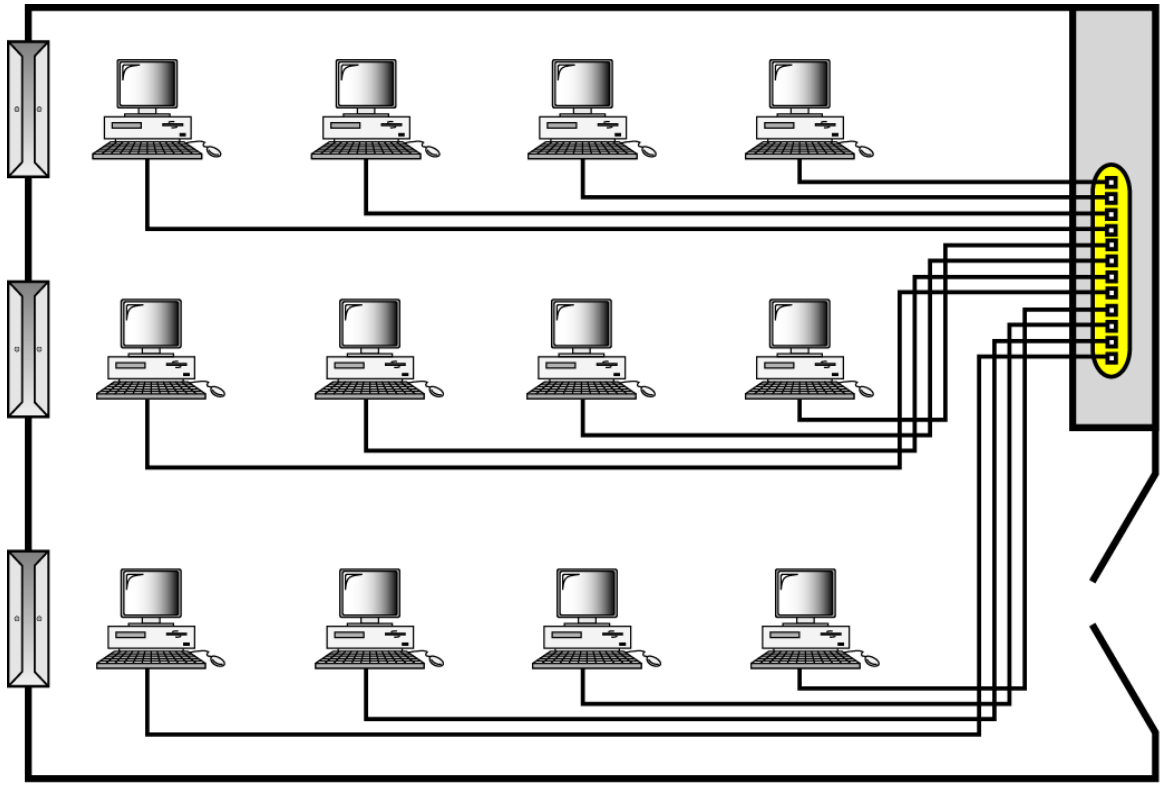
LAN/WLAN



MAN



WAN



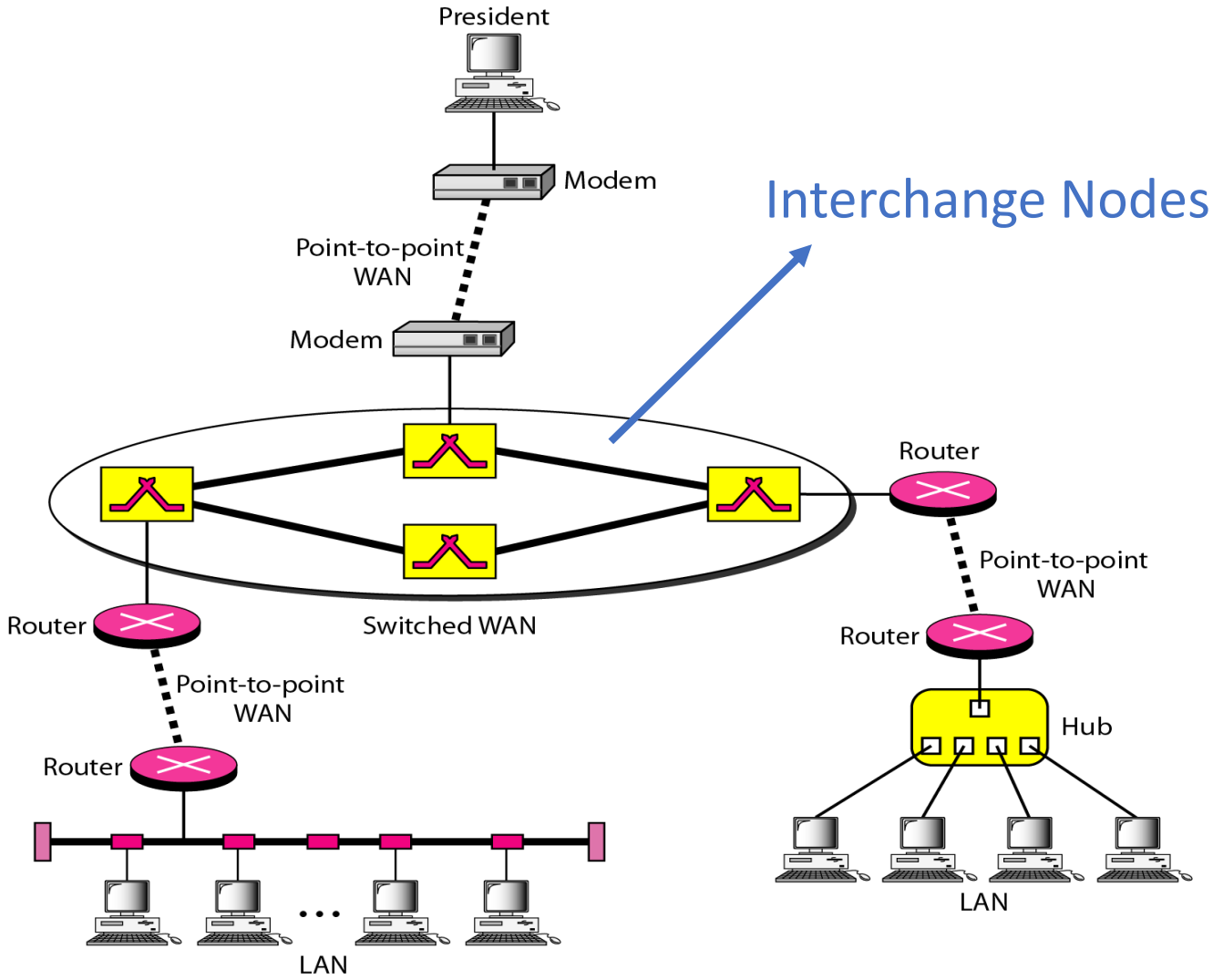
Example of LAN connecting 12 computers to a switch

Introduction

Complexity of WANs

- The complexity of the topology may increase with the increased dimension of the network.

- WANs connecting nations or continents can obviously be very complex and heterogeneous.



Example of heterogeneous WAN

Introduction

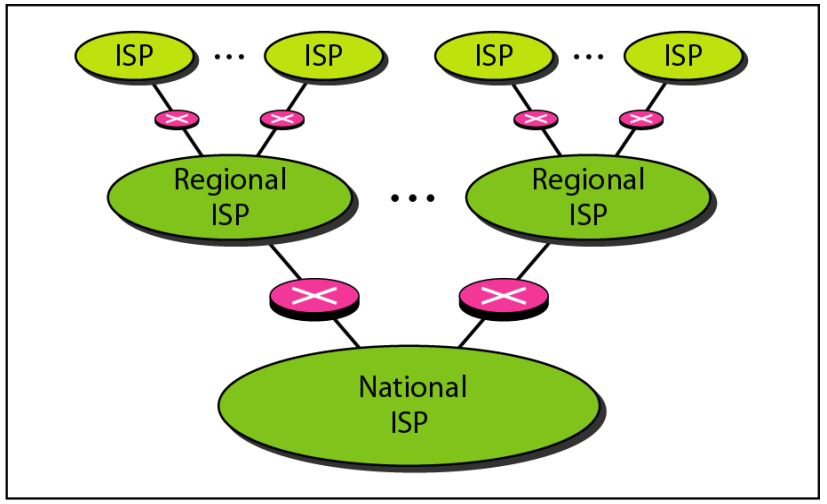
Service Providers

- The main role of computer networks is to allow data exchange between multiple devices from different locations, but raw data are just a mean to reach **services** and **resources** provided by the network.
- A network **resource** is a remote “object” we want to have access to (web pages, storage, computation, video or audio files, etc.).
- A network **service** is an “action” that a remote device performs for us (give me the current time, let me send an e-mail, let me store files, etc.).
- The difference between these two definitions is somehow subtle (e.g., let me *see* a *video*, *give* me access to a *web page*, are actions involving objects).
- Entities who offers services on Internet are called **Service Providers**.

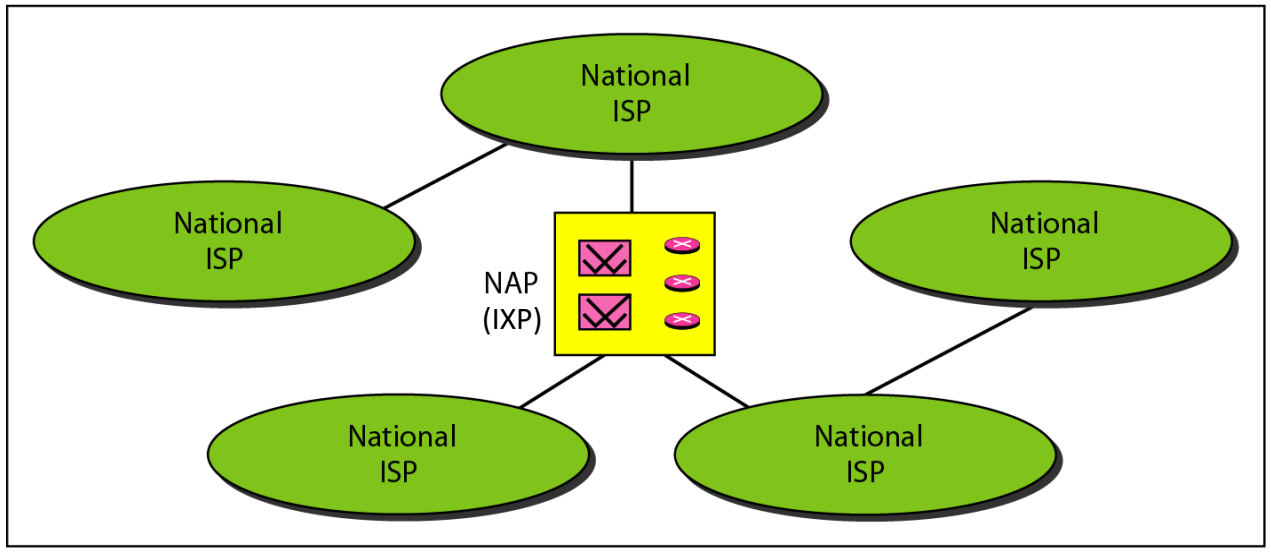
Introduction

Internet Service Providers

- **Internet access** is the basic service.
- An **Internet Service Provider** (ISP) is an organization that provides services for accessing, using, or participating the Internet.
- ISPs can be organized as commercial, community-owned, non-profit, or privately owned (e.g., by private companies or universities).
- Different ISPs exchange data through **Network (neutral) Access Points** (NAPs) or **Internet Exchange Points** (IXPs).



a. Structure of a national ISP

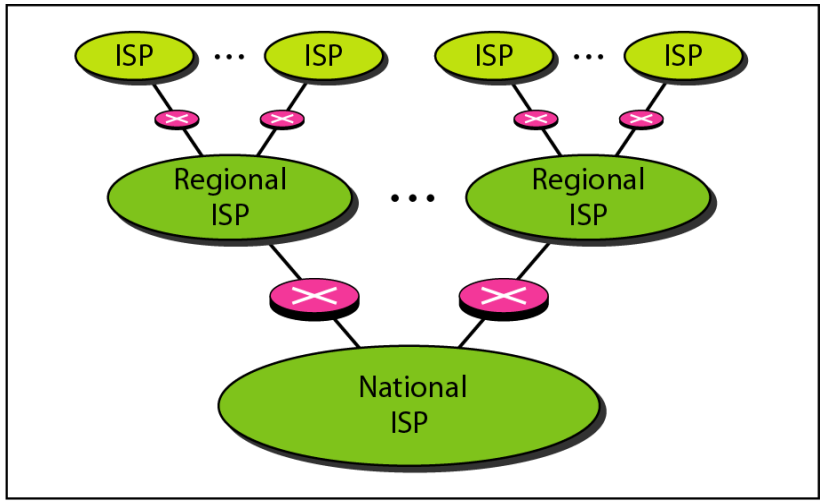


b. Interconnection of national ISPs

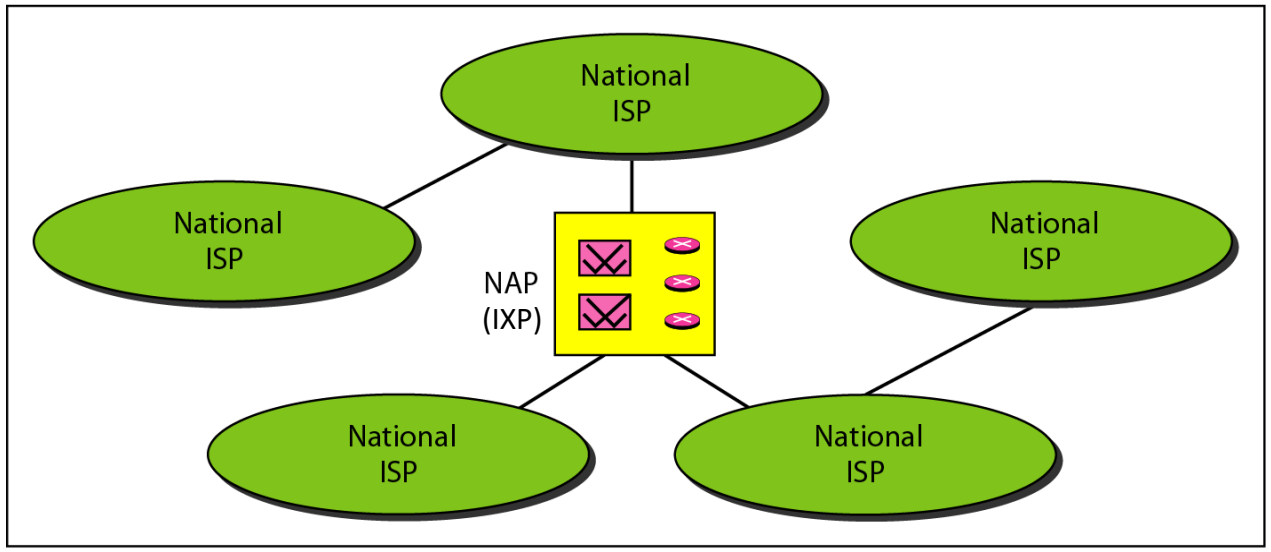
Introduction

Internet Service Providers

- An ISP network is hierarchically defined:
  - A **point of presence** (PoP) is a group of one or more routers used by ISPs to reach the customers.
  - **Access** ISPs for local areas.
  - **Regional** ISPs for larger areas.
  - **National** ISPs for nations.
- An **Internet exchange point** (IXP) works as a meeting point between multiple ISPs, typically managed by third parties.



a. Structure of a national ISP



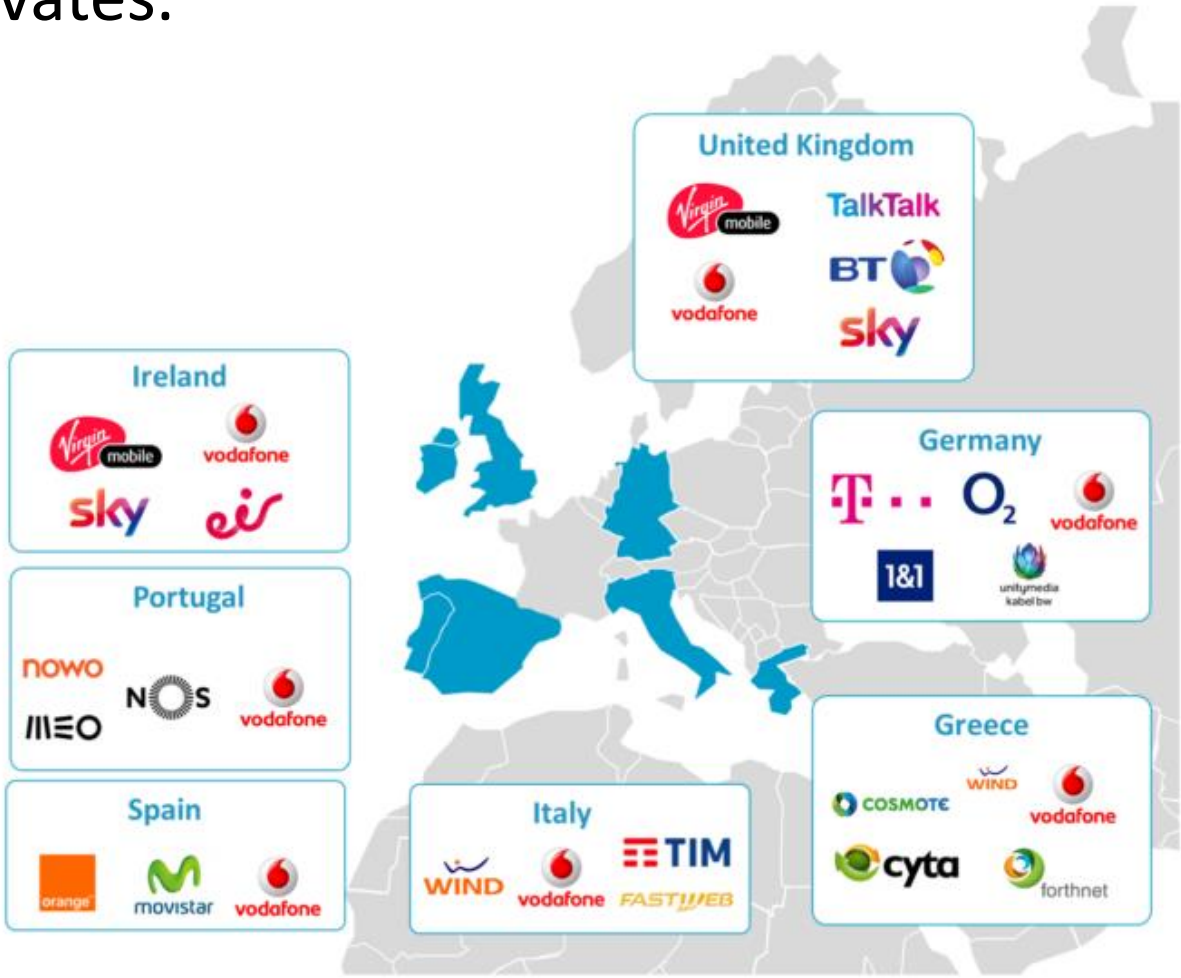
b. Interconnection of national ISPs

Introduction

Example of ISPs

- On a national scale there can be several commercial ISPs which provides internet services to companies and privates.

- Some Italian ISPs are:
  - Telecom Italia.
  - Fastweb.
  - Vodafone.
  - Tiscali.
  - …

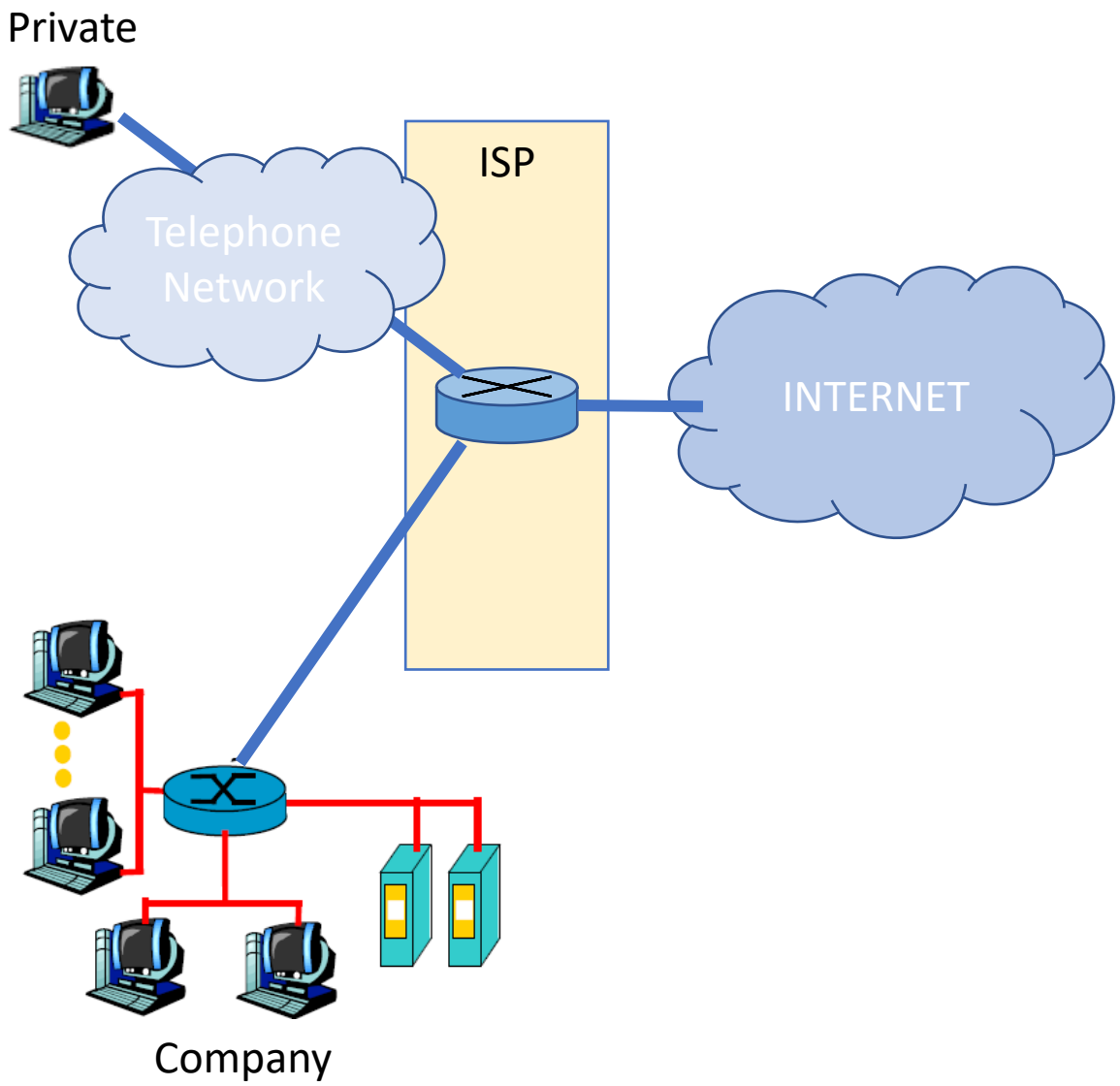


Introduction

Internet Connection

- Private internet connections are established via **Modem** enabled by the telephone network.
  - Analogic (56kbps).
  - Integrated Services Digital Network ISDN (128kbps).
  - Asymmetric Digital Subscriber Line ADSL (1Mbps to 20Mbps).
  - Twisted-Pair Copper Wire (10Mbps to 100Mbps).
  - Optical Fiber (50Mbps to 40Gbps).

- Companies (especially medium/large ones) can have a direct/dedicated connection to the ISP.



Introduction

Example of the GARR Network

- The internet access of Italian universities is managed by GARR.
- GARR (Gruppo per l'Armonizzazione delle Reti della Ricerca) is the Italian national computer network for universities and research.
- The GARR network is connected to other national research and education networks in Europe and the world, is an integral part of the global Internet.

